

II. Recognisability by Aperiodic Semigroups and First-Order Logic Definability

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Idempotents and Aperiodicity

$(S, \cdot)$ semigroup $s \cdot s = s$ idempotent

Example 1 Each element of $(\{1, 2, \dots, n\}, \max)$ is idempotent

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Idempotent Power Lemma Let $(S, \cdot)$ be a **finite** semigroup.
For each $s \in S$, the set $\{s, s^2, s^3, \dots\}$ contains a unique idempotent $s^!$.

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A finite semigroup $(S, \cdot)$ is **aperiodic** iff $s^! = s^! \cdot s$, for all $s \in S$

Example 2 $(\{1, 2, \dots, n\}, \max)$ is aperiodic

Example 3 $(\{0, 1\}, +_2)$ is periodic

Star-Free Expressions

$a \in \Sigma$	$\emptyset$	LK	$L \cup K$	$\neg L$
the language $\{a\}$	the empty language	concatenation	union	complement wrt Σ^*

Example 1

$$\neg \emptyset = \Sigma^*$$

Example 2

$$\neg \emptyset a \neg \emptyset = \Sigma^* a \Sigma^*$$

Example 3

$$\neg(\neg \emptyset a \neg \emptyset) = b^* \text{ assuming that } \Sigma = \{a, b\}$$

Example 4

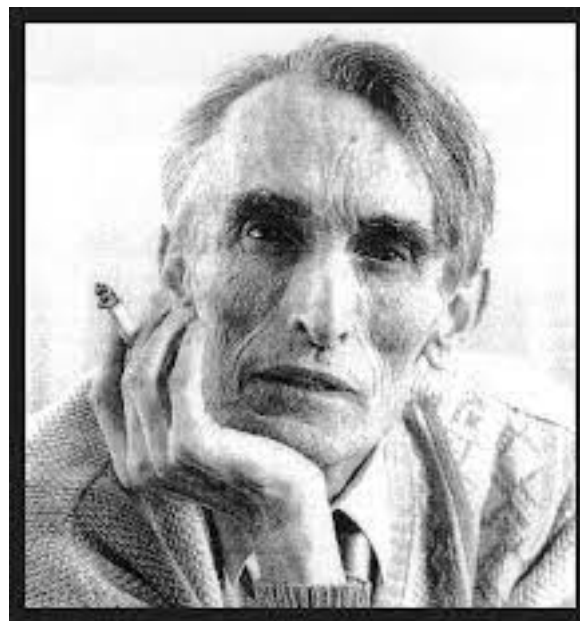
$$\neg(\neg \emptyset c \neg \emptyset) = \{a, b\}^* \text{ assuming that } \Sigma = \{a, b, c\}$$

Schützenberger-McNaughton-Papert

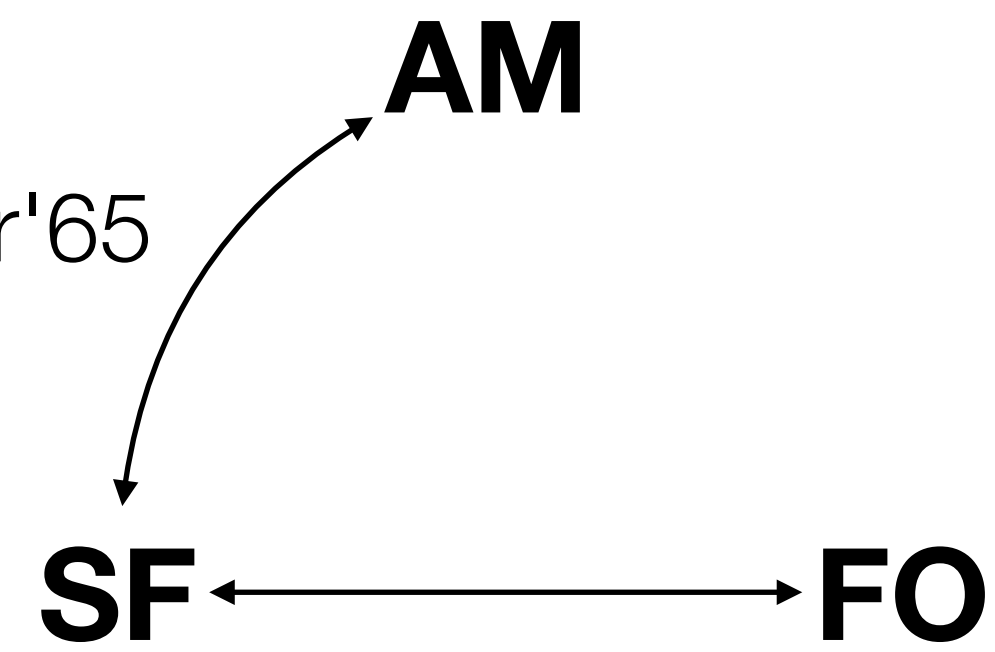
Theorem

The following are equivalent for $L \subseteq \Sigma^*$:

1. L is recognised by a finite aperiodic monoid
2. L is star-free
3. L is first-order definable



Schützenberger'65



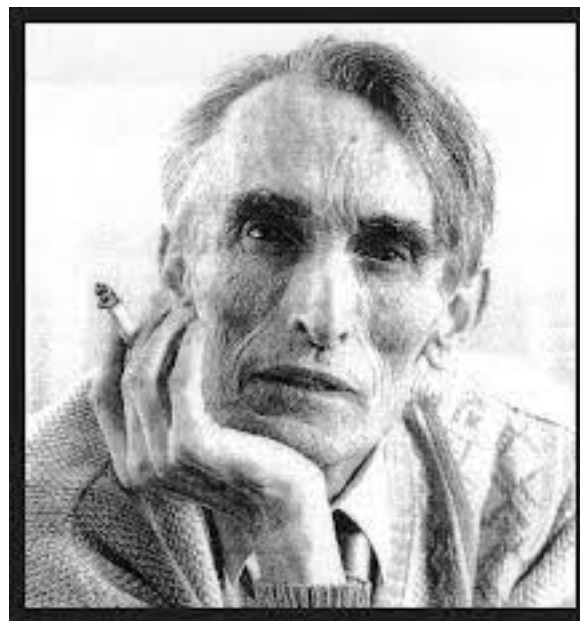
McNaughton, Papert'71

Schützenberger-McNaughton-Papert

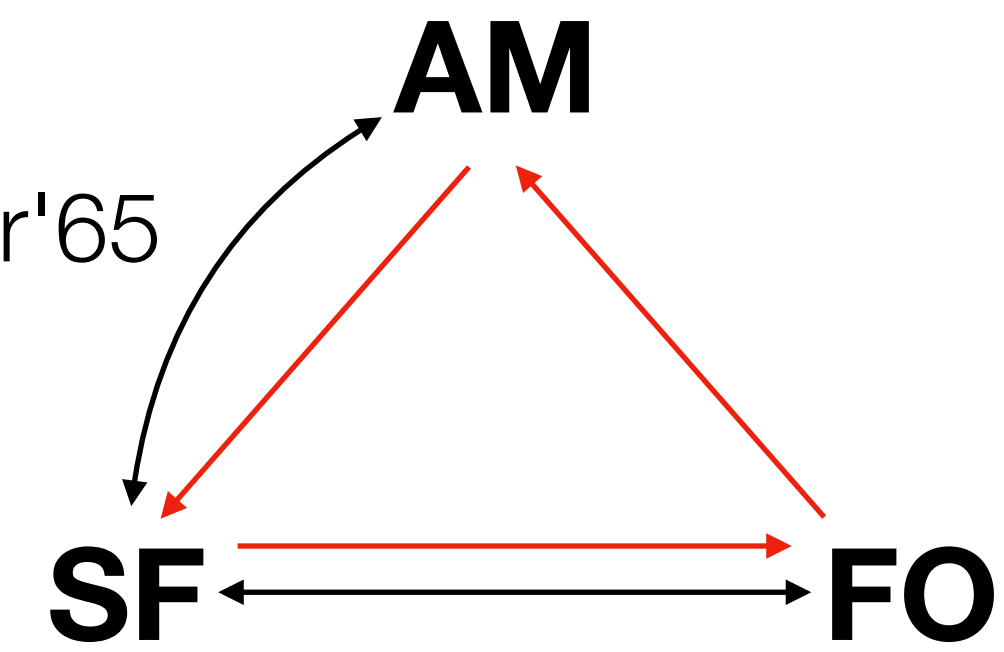
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Star-Free $\Rightarrow$ First-Order

$$a \quad \exists x. \textit{first}(x) \wedge \textit{last}(x) \wedge a(x)$$

$$\emptyset \quad \textit{false}$$

$$\text{LK} \quad \exists x \exists y \exists z. \textit{first}(x) \wedge \textit{last}(y) \wedge \underbrace{\varphi_{\text{L}}[x, z-1] \wedge \varphi_{\text{K}}[z, y]}_{\exists t. \psi \rightarrow \exists t. x \leq t < z \wedge \psi}$$

$$\text{L} \cup \text{K} \quad \varphi_{\text{L}} \vee \varphi_{\text{K}}$$

$$\neg \text{L} \quad \neg \varphi_{\text{L}}$$

First-Order $\Rightarrow$ Aperiodic Monoid

$$qr(a(x)) \stackrel{\text{def}}{=} qr(x \leq y) \stackrel{\text{def}}{=} 0$$

$$qr(\varphi \wedge \psi) \stackrel{\text{def}}{=} \max(qr(\varphi), qr(\psi))$$

$$qr(\neg \varphi) \stackrel{\text{def}}{=} qr(\varphi)$$

$$qr(\exists x. \varphi) \stackrel{\text{def}}{=} 1 + qr(\varphi)$$

$$\overbrace{\forall x . a(x) \Rightarrow (\underbrace{\exists y . y < x \wedge b(y)}_1) \wedge (\underbrace{\exists y . y > x \wedge b(y)}_1)}_2$$

$\textcolor{red}{W} \equiv_k \textcolor{blue}{W} \stackrel{\text{def}}{=} \textcolor{red}{W}$ and $\textcolor{blue}{W}$ satisfy the same FO sentences of quantifier rank $\leq k$

Characterisation of $\equiv_k$

$w \equiv_k w$ $\stackrel{\text{def}}{=} w$ and w satisfy the same FO sentences of quantifier rank $\leq k$

Lemma

$w \equiv_{k+1} w$ IFF $w \in LaK \Leftrightarrow w \in LaK$, for all $a \in \Sigma$ and $L, K \subseteq \Sigma^*$ equivalence classes of $\equiv_k$

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" $\Rightarrow$ " Each language $LaK \subseteq \Sigma^*$ is defined by a formula

$$\exists x . a(x) \wedge \overbrace{\varphi_L^{<x} \wedge \varphi_K^{>x}}^{\leq k+1}$$

$\underbrace{\varphi_L^{<x}}_{\leq k} \quad \underbrace{\varphi_K^{>x}}_{\leq k}$

φ_L and φ_R exist because there are finitely many equivalent formulae of quantifier rank $\leq k$

Ehrenfeucht-Fraïssé Games

Spoiler

$W, x_1, \dots, x_n$

Duplicator

$W, x_1, \dots, x_n$

Round

Ehrenfeucht-Fraïssé Games

Spoiler

$W, x_1, \dots, x_n \quad x_{n+1}$

Duplicator

$W, x_1, \dots, x_n \quad x_{n+1}$

Round

k

Ehrenfeucht-Fraïssé Games

Spoiler

$W, X_1, \dots, X_n \quad X_{n+1} \quad X_{n+2}$

Duplicator

$W, X_1, \dots, X_n$ X_{n+1} X_{n+2}

Round

$$k \quad k-1$$

Ehrenfeucht-Fraïssé Games

Spoiler

$W, x_1, \dots, x_n$ x_{n+1} x_{n+2} $\dots$ x_{n+k}

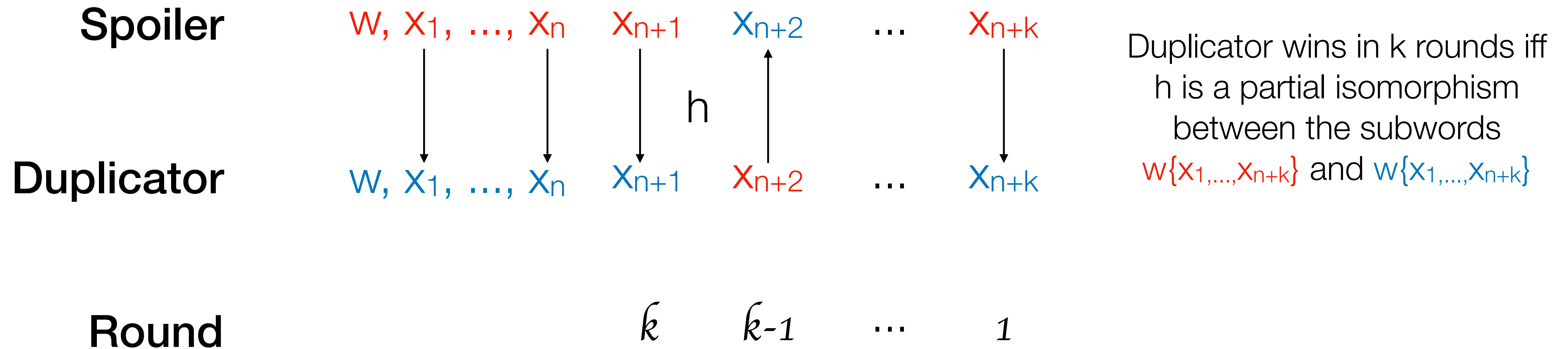
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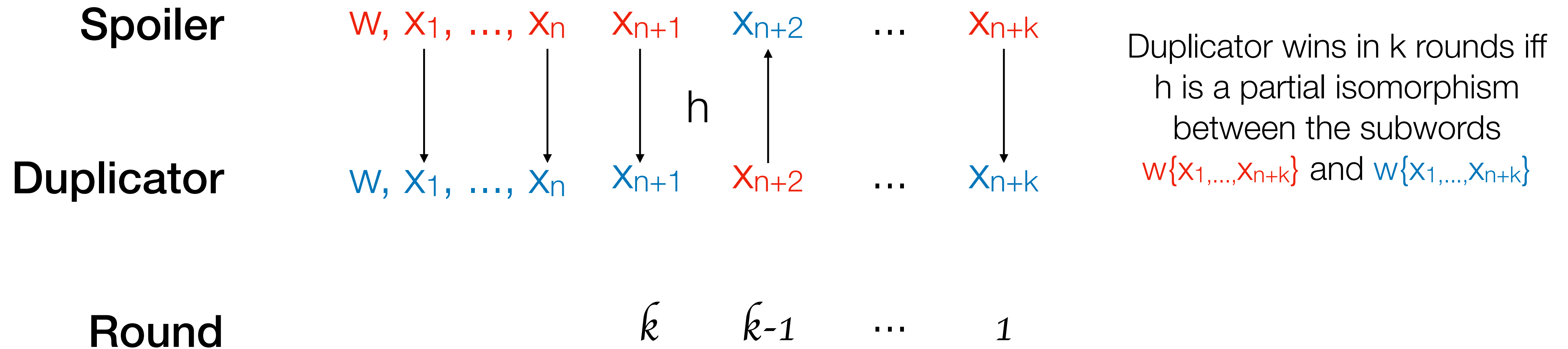
Round

k $k-1$ $\dots$ 1

Ehrenfeucht-Fraïssé Games



Ehrenfeucht-Fraïssé Games



Theorem

Duplicator has a winning strategy in k rounds from $W, X_1, \dots, X_n, W, X_1, \dots, X_n$

IFF $W, X_1, \dots, X_n \models \varphi \Leftrightarrow W, X_1, \dots, X_n \models \varphi$ for all FO-formulae φ having $qr(\varphi) \leq k$

Characterisation of $\equiv_k$

$w \equiv_k w \stackrel{\text{def}}{=} w$ and w satisfy the same FO sentences of quantifier rank $\leq k$

Lemma

$w \equiv_{k+1} w$ IFF $w \in LaK \Leftrightarrow w \in LaK$, for all $a \in \Sigma$ and $L, K \subseteq \Sigma^*$ equivalence classes of $\equiv_k$

" $\Leftarrow$ " $w \not\equiv_{k+1} w \Rightarrow$ there exists a sentence φ , $qr(\varphi) \leq k+1$, that distinguishes w from w

Assume w.l.o.g. that $\varphi \stackrel{\text{def}}{=} \exists x . \psi$ and $qr(\psi) \leq k$

Spoiler wins from w, x_1, w, x_1 if and only if either one of the below is true:

1. $w(x_1) \neq w(x_1)$
2. Spoiler has a winning strategy in k rounds on w before x_1 and on w before x_1
3. Spoiler has a winning strategy in k rounds on w after x_1 and on w after x_1

First-Order $\Rightarrow$ Aperiodic Monoid

L is the language of a FO sentence φ of quantifier rank k

$$\begin{array}{ccc} (\Sigma^*, \varepsilon, \cdot) & \xrightarrow{h_k(w) \stackrel{\text{def}}{=} [w]_{/\equiv_k}} & (\Sigma^*_{/\equiv_k}, [\varepsilon]_{/\equiv_k}, \circ) \\ \text{IU} & & \text{IU} \\ L & = & h_k^{-1}(\{[w]_{/\equiv_k} \mid w \models \varphi\}) \end{array}$$

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 \text{IU} & & \text{IU} \\
 L & = & h_k^{-1}(\{[w]_{/\equiv_k} \mid w \models \varphi\})
 \end{array}$$

- ▶ $h_k(w_1 w_2)$ is computed from $h_k(w_1)$ and $h_k(w_2)$ as $h_k(w_1) \circ h_k(w_2)$
- ▶ $(h_k(w_1) \circ h_k(w_2)) \circ h_k(w_3) = h_k(w_1 w_2 w_3) = h_k(w_1) \circ (h_k(w_2) \circ h_k(w_3))$
- ▶ $h_k(w) \circ h_k(\varepsilon) = h_k(w\varepsilon) = h_k(w)$
- ▶ $(\Sigma^*_{/\equiv_k}, [\varepsilon]_{/\equiv_k}, \circ)$ is aperiodic: $w^{2^k} \equiv_k w^{2^k+1}$, for all $w \in \Sigma^*$

Aperiodic Monoid $\Rightarrow$ Star-Free

$$\begin{array}{ccc}
 (\Sigma^*, \varepsilon, \cdot) & \xrightarrow{h} & (M, 1, \circ) \quad \text{finite aperiodic monoid} \\
 \text{IU} & & \text{IU} \\
 L & = & h^{-1}(N)
 \end{array}$$

We prove that $h^{-1}(m)$ is star-free, for each $m \in M$

Induction on $r(m) \stackrel{\text{def}}{=} ||M - MmM||$

The Simplification Lemma

$(M, 1, \circ)$ finite aperiodic monoid and $m \in M$

Lemma 1 (Simplification)

For all $p, q, r \in M$, $pqr=q \Rightarrow pq = q = qr$

Lemma 2

1. $M \subseteq Mm \Rightarrow m=1$

2. $M \subseteq MmM \Rightarrow m=1$

The Base Case

$$r(m)=0 \Rightarrow M = MmM \Rightarrow m=1 \quad \textbf{(Lemma 2)}$$

$$h^{-1}(1)=\underbrace{\{a \in \Sigma \mid h(a)=1\}}_{B \subseteq \Sigma}^*$$

$$B^* = \neg(\Sigma^* \neg B \Sigma^*) \text{ is star-free}$$

The Base Case

$$r(m)=0 \Rightarrow M = MmM \Rightarrow m=1 \quad \textbf{(Lemma 2)}$$

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$$B^* = \neg(\Sigma^* \neg B \Sigma^*) \text{ is star-free}$$

$$\left. \begin{array}{l} h(ua)=1 \\ h(u)1h(a)=1 \\ 1h(a)=1 \\ h(a)=1 \end{array} \right\} \textbf{SL}$$

Green Relations

m is a **prefix** of $n \stackrel{\text{def}}{=} mx = n$, for some $x \in M \stackrel{\text{def}}{=} mM \supseteq nM$

$$m \mathcal{R} n \stackrel{\text{def}}{=} mM = nM$$

m is a **suffix** of $n \stackrel{\text{def}}{=} xm = n$, for some $x \in M \stackrel{\text{def}}{=} Mm \supseteq Mn$

$$m \mathcal{L} n \stackrel{\text{def}}{=} Mm = Mn$$

m is an **infix** of $n \stackrel{\text{def}}{=} xmy = n$, for some $x, y \in M \stackrel{\text{def}}{=} MmM \supseteq MnM$

$$m \mathcal{J} n \stackrel{\text{def}}{=} MmM = MnM$$

The Inductive Step

$$m \neq 1 \Rightarrow h^{-1}(m) = (U\Sigma^* \cap \Sigma^*V) \setminus (\Sigma^*C\Sigma^* \cup \Sigma^*W\Sigma^*)$$

$$U \stackrel{\text{def}}{=} \bigcup_{(n,a) \in E} h^{-1}(n)a\Sigma^*$$

$$E \stackrel{\text{def}}{=} \{(n,a) \in M \times \Sigma \mid nh(a) \mathcal{R} m, n \notin mM\}$$

$$V \stackrel{\text{def}}{=} \bigcup_{(a,n) \in F} \Sigma^*ah^{-1}(n)$$

$$F \stackrel{\text{def}}{=} \{(a,n) \in \Sigma \times M \mid m \mathcal{L} h(a)n, n \notin Mm\}$$

$$C \stackrel{\text{def}}{=} \{a \in \Sigma \mid m \notin Mh(a)M\}$$

$$W \stackrel{\text{def}}{=} \bigcup_{(a,n,b) \in G} \Sigma^*ah^{-1}(n)b\Sigma^*$$

$$G \stackrel{\text{def}}{=} \{(a,n,b) \in \Sigma \times M \times \Sigma \mid m \in (Mh(a)nM \cap Mnh(b))\}$$

It suffices to prove that $U, V, W \subseteq \Sigma^*$ are star-free languages
 ($C \subseteq \Sigma \Rightarrow \Sigma^*C\Sigma^*$ is star-free)

The Inductive Step

$$m \neq 1 \Rightarrow h^{-1}(m) \subseteq U\Sigma^* \quad (h^{-1}(m) \subseteq \Sigma^*V \text{ is symmetric})$$

$$U \stackrel{\text{def}}{=} \bigcup_{(n,a) \in E} h^{-1}(n)a\Sigma^*$$

$$E \stackrel{\text{def}}{=} \{(n,a) \in M \times \Sigma \mid nh(a) \mathcal{R} m, n \notin mM\}$$

Let $u = px \in h^{-1}(m)$ where p is the shortest such that $h(p) \mathcal{R} m$ (clearly $h(u) \mathcal{R} m$)

$$p = \varepsilon \Rightarrow 1 \mathcal{R} m \Rightarrow M = Mm \Rightarrow m = 1 \times \textbf{(Lemma 2)}$$

$$p = ra \Rightarrow \underbrace{h(r)}_{\stackrel{\text{def}}{=} n} h(a) \mathcal{R} m \text{ and } \neg \underbrace{n \mathcal{R} m}_{\text{choice of } p} \Rightarrow (n,a) \in E$$

$$u = px = rax \in h^{-1}(n)a\Sigma^* \subseteq U\Sigma^*$$

The Inductive Step

$$\boxed{h^{-1}(m) \cap \Sigma^* C \Sigma^* = \emptyset}$$

$$C \stackrel{\text{def}}{=} \{a \in \Sigma \mid m \notin M h(a) M\}$$

Let $u \in \Sigma^*$ be such that $h(u)=m$

$u \in \Sigma^* C \Sigma^* \Rightarrow m \in M h(a) M$, for some $a \in C$ ✕

The Inductive Step

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$u \in \Sigma^* C \Sigma^* \Rightarrow m \in M h(a) M$, for some $a \in C$ ✕

$$\boxed{h^{-1}(m) \cap \Sigma^* W \Sigma^* = \emptyset}$$

$$W \stackrel{\text{def}}{=} \bigcup_{(a,n,b) \in G} \Sigma^* a h^{-1}(n) b \Sigma^*$$

$$G \stackrel{\text{def}}{=} \{(a, n, b) \in \Sigma \times M \times \Sigma \mid m \in (Mh(a)nh(b)M) \setminus Mh(a)nh(b)M\}$$

$u \in \Sigma^* W \Sigma^* \Rightarrow u \in \Sigma^* a h^{-1}(n) b \Sigma^*$ for some $(a,n,b) \in G \Rightarrow m \in M h(a) nh(b) M$ ✕

Ideal Decomposition

$(M, 1, \circ)$ finite aperiodic monoid and $m \in M$

Lemma 3 (Ideal Decomposition)

$\{m\} = (mM \cap Mm) \setminus J_m$, where $J_m \stackrel{\text{def}}{=} \{s \in M \mid m \notin MsM\}$

$\{m\} \subseteq (mM \cap Mm) \setminus J_m$ is trivial

Let $s \in (mM \cap Mm) \setminus J_m \Rightarrow s = mp = rm$, for some $p, r \in M$

$m = usv$, for some $u, v \in M$

$$m = usv = u(rm)v = \underbrace{(ur)m}_{(SL)}v = urm = us = \underbrace{ump}_{(SL)} = mp = s$$

$$(mM \cap Mm) \setminus J_m \subseteq \{m\}$$

The Inductive Step

$$h^{-1}(m) \supseteq (U\Sigma^* \cap \Sigma^*V) \setminus (\Sigma^*C\Sigma^* \cup \Sigma^*W\Sigma^*)$$

Let $u \in (U\Sigma^* \cap \Sigma^*V) \setminus (\Sigma^*C\Sigma^* \cup \Sigma^*W\Sigma^*)$ and $s \stackrel{\text{def}}{=} h(u)$

We prove $\{s\} \subseteq (mM \cap Mm) \setminus J_m = \{m\}$, where $J_m \stackrel{\text{def}}{=} \{s \in M \mid m \notin MsM\}$ **(ID)**

$u \in U\Sigma^* \Rightarrow u \in h^{-1}(n)a\Sigma^*$, for some $(n,a) \in E \Rightarrow s \in \underbrace{nh(a)M}_{nh(a) \mathcal{R} m} \Rightarrow s \in mM$

$$U \stackrel{\text{def}}{=} \bigcup_{(n,a) \in E} h^{-1}(n)a\Sigma^*$$

$$E \stackrel{\text{def}}{=} \{(n,a) \in M \times \Sigma \mid nh(a) \mathcal{R} m, n \notin mM\}$$

$u \in \Sigma^*V \Rightarrow s \in Mm$ is symmetric

The Inductive Step

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$$s \notin J_m \Leftrightarrow m \in MsM$$

Let f be a least factor of u such that $m \notin Mh(f)M$

► $f = \varepsilon \Rightarrow m \notin M \times$

► $f \in \Sigma \Rightarrow u \in \Sigma^*C\Sigma^* \times$

$$C \stackrel{\text{def}}{=} \{a \in \Sigma \mid m \notin Mh(a)M\}$$

The Inductive Step

$$h^{-1}(m) \supseteq (U\Sigma^* \cap \Sigma^*V) \setminus (\Sigma^*C\Sigma^* \cup \Sigma^*W\Sigma^*)$$

Let $u \in (U\Sigma^* \cap \Sigma^*V) \setminus (\Sigma^*C\Sigma^* \cup \Sigma^*W\Sigma^*)$ and $s \stackrel{\text{def}}{=} h(u)$

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$s \notin J_m \Leftrightarrow m \in MsM$

Let f be a least factor of u such that $m \notin Mh(f)M$

► $f = agb$, for some $a, b \in \Sigma$ and $g \in \Sigma^* \Rightarrow h(f) = h(a) \underbrace{h(g)}_{\stackrel{\text{def}}{=} n} h(b)$

$\Rightarrow m \in Mh(a) \underbrace{nM \cap Mnh(b)}_{\text{choice of } f} M \setminus Mh(a)nh(b)M \Rightarrow (a, n, b) \in G \Rightarrow f \in W \Rightarrow u \in \Sigma^*W\Sigma^* \times$

$$W \stackrel{\text{def}}{=} \bigcup_{(a,n,b) \in G} \Sigma^* a h^{-1}(n) b \Sigma^*$$

$$G \stackrel{\text{def}}{=} \{(a, n, b) \in \Sigma \times M \times \Sigma \mid m \in (Mh(a)nM \cap Mnh(b)M) \setminus Mh(a)nh(b)M\}$$

The Eggbox Lemma

Lemma 4 (Eggbox) m prefix of n and $m \mathcal{J} n \Rightarrow m \mathcal{R} n$

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Lemma 4 (Eggbox) m prefix of n and $m \mathcal{J} n \Rightarrow m \mathcal{R} n$

$MmM = MnM \Rightarrow m = xny$, for some $x, y \in M$

$mM \supseteq nM \Rightarrow n = mz$, for some $z \in M$

$m = xny = x \underbrace{mzy}_{(SL)} = mzy \Rightarrow mM = mzyM \subseteq mzM = nM \Rightarrow mM = nM$

The Inductive Step

U is star-free (symmetric proof for V)

$$U \stackrel{\text{def}}{=} \bigcup_{(n,a) \in E} h^{-1}(n)a\Sigma^*$$

$$E \stackrel{\text{def}}{=} \{(n,a) \in M \times \Sigma \mid nh(a) \mathcal{R} m, n \notin mM\}$$

$$(n,a) \in E \Rightarrow nh(a) \mathcal{R} m \Rightarrow nh(a)M = mM \Rightarrow Mnh(a)M = MmM \subseteq MnM \Rightarrow r(n) \leq r(m)$$

$$\left. \begin{array}{l} \text{Suppose } r(n)=r(m) \Rightarrow n \mathcal{J} m \\ nh(a) \mathcal{R} m \Rightarrow mM = nh(a)M \subseteq nM \Rightarrow m \text{ prefix of } n \end{array} \right\} \textbf{(EL)} \Rightarrow n \mathcal{R} m, \nexists n \notin mM$$

Then $r(n) < r(m) \Rightarrow h^{-1}(n)$ is star-free **(IH)** $\Rightarrow U$ is star-free

The Inductive Step

W is star-free

$$W \stackrel{\text{def}}{=} \bigcup_{(a,n,b) \in G} \Sigma^* a h^{-1}(n) b \Sigma^*$$

$$G \stackrel{\text{def}}{=} \{(a, n, b) \in \Sigma \times M \times \Sigma \mid m \in (M h(a) n M \cap M n h(b) M) \setminus M h(a) n h(b) M\}$$

$$(a,n,b) \in G \Rightarrow m \in M n h(b) M \subseteq M n M \Rightarrow M m M \subseteq M n M \Rightarrow r(n) \leq r(m)$$

$$\left. \begin{array}{l} \text{Suppose } r(n)=r(m) \Rightarrow n \mathcal{J} m \Rightarrow n \in M m M \\ m \in M n h(b) M \end{array} \right\} \Rightarrow n \in M n h(b) M \Rightarrow n = u n h(b) \underbrace{v = n h(b) v}_{\textbf{(SL)}}$$

$$m \in M h(a) n M \Rightarrow m = x h(a) n y = x h(a) n h(b) v y, \nexists m \notin M h(a) n h(b) M$$

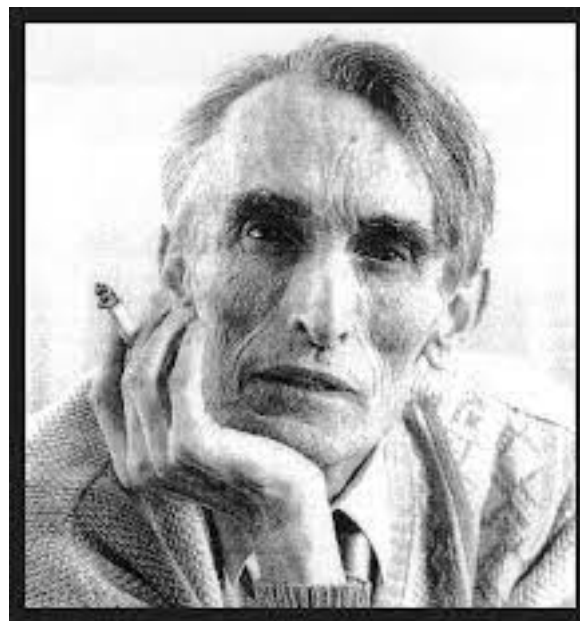
Then $r(n) < r(m) \Rightarrow h^{-1}(n)$ is star-free **(IH)** $\Rightarrow W$ is star-free

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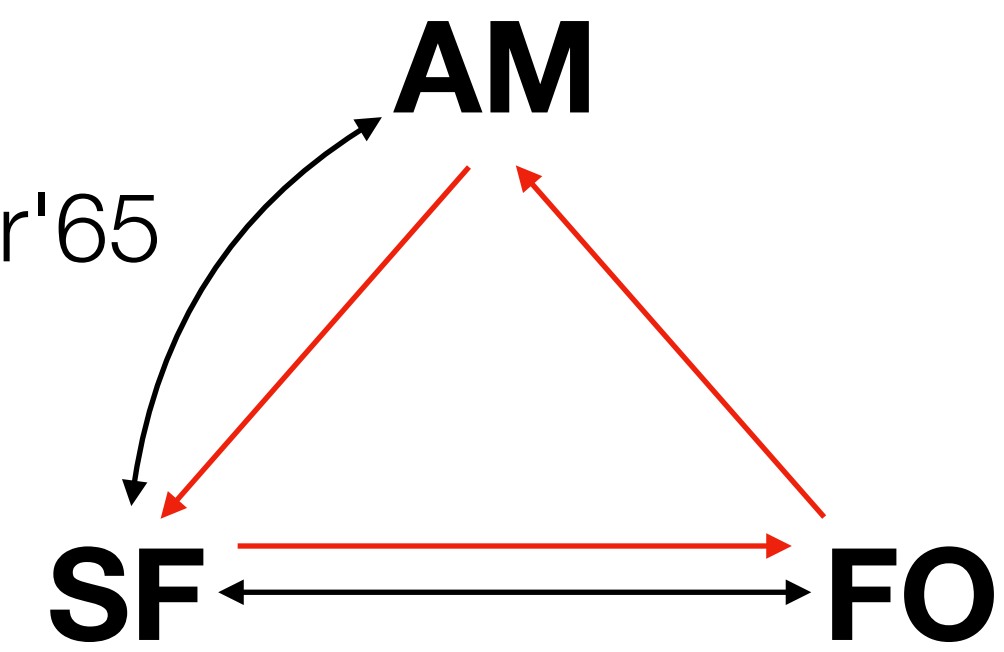
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